

TensorForge 2.0 / Phase 2 MVP

Support ticket classification and routing

Team Fade | Final technical report | 9 October 2026

1. Solution and design decisions

TensorForge classifies RideEat support tickets, a fictional ride-hailing and food-delivery platform. For each ticket it assigns one of 11 primary categories, maps that category to the organizer-defined team, returns an optional secondary category, predicts urgency, and returns a confidence with a human-review flag when confidence is below 0.5. It accepts email, chat, and call transcripts in English, Sinhala, Tamil, and transliterated or mixed text. No generative LLM or external classification API is called at inference.

The deployed model fuses a self-trained word and character TF-IDF classifier with a fine-tuned multilingual encoder. The development-validation macro-F1 is 0.8022, compared with 0.6558 for the selected classical baseline. The model version is v1.0.0-38ecb9bd. This 0.8022 figure is development validation, not an untouched holdout: fusion weights, temperatures, and thresholds were selected using out-of-fold predictions and labels across all 4,800 tickets, including the validation labels.

Decision	Reason
Classical + encoder fusion	Character n-grams cover spelling; the encoder adds multilingual meaning.
Macro-F1 for selection	Classes range from about 3% to 18%, so accuracy would hide rare classes.
Contract middleware	Auth runs before parsing. Errors stay JSON with the required status codes.
CPU-only offline image	Int8 ONNX and local files. The image serves with networking disabled.
One worker + SQLite jobs	Large jobs poll from durable state and are marked interrupted after restart.

Live service: <https://tensorforge-fade.southindia.cloudapp.azure.com> | Demo: /demo/ | Repository: github.com/batman2400/Tensor-forge | Release: v1.0.1

2. Data, training, and scores

The organizer set has 4,000 training tickets and 800 validation tickets. Across all 4,800, secondary labels and urgent tickets are each about 10%. Languages are English 35%, Singlish 25%, Sinhala 20%, Tamil 15%, mixed 2.5%, and Tanglish 2.5%. Channels are chat 45%, email 30%, and call transcript 25%. ticket_id and the language label are not model features. Text is NFC-normalized. The API allows 10,000 characters.

The classical branch is svc_word_char (word and character TF-IDF, linear SVM). Logistic regression, SGD, naive Bayes, and LightGBM were compared on the same split. The encoder is intfloat/multilingual-e5-small (MIT), fine-tuned with task heads and exported to int8 ONNX. The serving encoder was refit on all 4,800 tickets for four epochs. Fusion gives the encoder 0.75 of the category and secondary vote and 0.50 of the urgency vote.

Configuration	Accuracy	Macro-F1	Urgent F1	Secondary exact
Word TF-IDF logistic regression	0.4600	0.5014	0.8252	0.9525
Word + char logistic regression	0.5787	0.6037	0.8725	0.9513
Word + char SVC baseline	0.6338	0.6558	0.8295	0.9550
SVC + encoder fusion	0.7975	0.8022	0.9080	0.9637

Recorded fusion ECE is 0.0589. Sinhala is the weakest language slice (macro-F1 0.624 on 160 tickets). Mixed and Tanglish each have only 20 tickets. A score of the final all-data encoder on the same 800 rows is in-sample; an API check of those rows scored about 0.994 macro-F1 and is not validation accuracy. Pooled random-fold scores are higher because tickets share opening templates. Leave-one-language-out retraining was not run. An untouched organizer holdout remains the decisive generalization test.

3. Container evidence (9 October 2026)

Image sha256:45131d2759b9a58d5d40529f5c740f2ee7c5ceb7464eef1b0a80d793dc31a3db (983 MB), built from commit 52e612b5fccbe228bfc2bb3520df1427726845a9 with Git LFS weights hydrated. The image label records that commit and version v1.0.1. A scan of the image filesystem found no `tf2_key` and no `API_KEY`. The base Python image sets `GPG_KEY` to the public CPython release signing key; that is not a project secret. The service key is supplied only at runtime.

Check under 2 CPU / 4 GB, swap capped at 4 GB	Result
Startup to HTTP 200 /health	4.3 s (limit 120 s)
Startup with <code>--network none</code>	3.3 s, health 200
Contract, auth, 100-ticket batch, jobs	256/256 checks passed
2,000-ticket job / poll p95	33 s / 0.02 s
5,000-ticket job / poll p95	80 s / 0.02 s; submit under 5 s
Peak resident memory during those jobs	572 MiB
Restart during a 1,500-ticket job	failed / interrupted; health in 3.1 s

GitHub Actions run 37921671034 on commit 52e612b succeeded for both the hygiene job and the Linux test job (LFS checkout, non-slow pytest). The in-sample 0.994 figure above came from this same container run.

4. Hosted service

The tested image was loaded on the South India VM and the container was recreated with restart policy unless-stopped, 2 CPUs, and a 3 GB memory cap. The VM has 3.8 GB of RAM, so the live cap is 3 GB rather than the 4 GB proof above. Job storage stayed on the Docker volume `tf_data`. The previous image sha256:e18b0c716e0240c4dc86f9a04703380e9a82cd2a604856e4d30d3e21da88a11c remains on the VM as the rollback image. docker image inspect on the VM reports the same digest as the local build. Public HTTPS /health returned 200 with `model_version v1.0.0-38ecb9bd`.

Earlier laptop failures were the uplink, not the application. From this network a 30 MB upload took about 81 s, so a 60 s client write timeout fired before the JSON 401 could be read. On the VM, through Caddy, the deployed image returned JSON 401 for a 30 MB unauthenticated upload in 0.15 s and JSON 413 for an authenticated 6 MB batch. Direct container checks on the VM were the same shape, in under 0.03 s.

Latency was measured again on the VM, through public HTTPS, after this image was deployed. Single-prediction p95 was 0.035 s (40 requests). A 100-ticket batch p95 was 1.53 s (5 requests). The 2,000-ticket job was accepted in 0.017 s and finished in 48.5 s. The 5,000-ticket job was accepted in 0.031 s and finished in 119.2 s. Poll p95 during that job was 0.012 s, and /health p95 was 0.010 s. All of those are under the one-second poll target and the five-second submit target. CPU credits remaining before the run were about 921.

Auto-shutdown: no DevTestLab shutdown schedule exists on the resource group. SSH is allowed from one operator address only. Ports 80 and 443 are open. Caddy terminates TLS and proxies to 127.0.0.1:8000 with a 64 MB body limit, above the application's limits, and a 120 s response header timeout. An Azure Monitor alert on `VmAvailabilityMetric` notifies the subscription operator.

5. Release identity and what is still open

Tag v1.0.0 still points at f164ba49b01626e8926335e46639fd2655fc076e, which is before the artifact-byte fix. It was not moved. This submission is release v1.0.1. The image digest to submit is sha256:45131d2759b9a58d5d40529f5c740f2ee7c5ceb7464eef1b0a80d793dc31a3db. An older plan entry, sha256:5d64978226bc891c3d7b09abf12ec19eb25d0c16aa8e523b6094d284b0639920, is not the running image. The fresh-clone hash mismatch described in an earlier draft of this report was fixed before commit cc6da4f; current artifact hashes match the manifest, and the image build runs `Engine.load()` before it succeeds.

Authenticated hosted checks in this session succeeded, including the upload cases above. The video is not in this report: it had not been recorded at publication time. The script in docs/video_script.md uses synthetic tickets in docs/demo_tickets.csv, tells the presenter to keep the API key off camera, and calls 0.802 development validation rather than an untouched holdout. The presenter still has to record about 15 minutes, upload it unlisted, and confirm playback while logged out of YouTube.

Hosting-credit balance was not visible to the Azure login used for this deploy, so remaining credit is not stated as a number here. The VM was running, CPU credits were nearly full, and disk use was 6.2 GB of 29 GB.